# Evaluation runs: replog-co (replicated-log-esop20)

generated 2026-11-03 from brain-data 48587ddd0 2026-11-03 23:12:27 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 8 | conforms | 0.0 | 4 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-03 |
| 2 | 16 | 128 | conforms | 0.0 | 2 | 1 h, 250 GB | 4 | 28 | brain03, 2026-11-03 |
| 3 | 144 | 4608 | conforms | 0.6 | 4 | 1 h, 250 GB | 36 | 2556 | brain03, 2026-11-03 |
| 4 | 2304 | 294912 | conforms | 48.1 | 24 | 1 h, 250 GB | 577 | 663000 | brain03, 2026-11-03 |
| 5 | 57600 | 29491200 | conforms | 17518.0 | 118 | 48 h, 250 GB | 14905 | 423756076 | brain04, 2026-11-01 |

### stateless: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 8 | conforms | 0.0 | 4 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-03 |
| 2 | 4 | 128 | conforms | 0.0 | 4 | 1 h, 250 GB | 4 | 10 | brain03, 2026-11-03 |
| 3 | 36 | 4608 | conforms | 0.6 | 4 | 1 h, 250 GB | 36 | 666 | brain03, 2026-11-03 |
| 4 | 576 | 294912 | conforms | 26.0 | 25 | 1 h, 250 GB | 576 | 166176 | brain03, 2026-11-03 |
| 5 | 14400 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair **early-reply**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 32 | 24 | violation | 0.0 | 4 | 1 h, 250 GB | 25 | 97 | brain03, 2026-11-03 |
| 2 | 16 | 512 | 384 | violation | 0.4 | 4 | 1 h, 250 GB | 388 | 6172 | brain03, 2026-11-03 |
| 3 | 144 | 18432 | 13824 | violation | 70.9 | 17 | 1 h, 250 GB | 13860 | 1993212 | brain03, 2026-11-03 |
| 4 | 2304 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair **early-reply**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 32 | 24 | violation | 0.0 | 4 | 1 h, 250 GB | 25 | 25 | brain03, 2026-11-03 |
| 2 | 4 | 512 | 384 | violation | 0.2 | 6 | 1 h, 250 GB | 388 | 1546 | brain03, 2026-11-03 |
| 3 | 36 | 18432 | 13824 | violation | 27.0 | 16 | 1 h, 250 GB | 13860 | 498330 | brain03, 2026-11-03 |
| 4 | 576 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

**Not run (stateless).** skipped: a smaller size had reached its limit: 5 early-reply fifo, 5 early-reply bag, 6 correct bag, 6 early-reply bag, 6 correct fifo, 6 early-reply fifo, 7 correct bag, 7 early-reply bag, 7 early-reply fifo, 7 correct fifo, 8 correct fifo, 8 early-reply bag, 8 early-reply fifo, 8 correct bag, 9 early-reply bag, 9 correct fifo, 9 early-reply fifo, 9 correct bag.

planned but never scheduled (grid points beyond the ladder): 5 early-reply fifo, 5 early-reply bag, 5 correct bag, 5 correct fifo, 6 correct bag, 6 early-reply bag, 7 correct bag.

# stateful

## stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1 | 4 | 8 | conforms | 0.6 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2 | 16 | 128 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 144 | 4608 | conforms | 0.6 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 2304 | 294912 | conforms | 21.2 | 26 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | 57600 | 29491200 | conforms | 599.7 | 122 | 48 h, 250 GB | brain05, 2026-11-02 |
| 6 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

## stateful: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 8 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2 | 4 | 128 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 36 | 4608 | conforms | 0.6 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 576 | 294912 | conforms | 20.7 | 27 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | 14400 | 29491200 | conforms | 1049.8 | 116 | 1 h, 250 GB | brain02, 2026-11-03 |
| 6 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-03 |

## stateful: pair **early-reply**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 32 | 24 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2 | 16 | 512 | 384 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 144 | 18432 | 13824 | violation | 1.9 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 2304 | 1179648 | 884736 | violation | 49.4 | 30 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | 57600 | 117964800 | 88473600 | violation | 2064.1 | 621 | 48 h, 250 GB | brain05, 2026-11-02 |

## stateful: pair **early-reply**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 32 | 24 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2 | 4 | 512 | 384 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 36 | 18432 | 13824 | violation | 2.0 | 6 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 576 | 1179648 | 884736 | violation | 49.8 | 30 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | 14400 | 117964800 | 88473600 | violation | 3571.1 | 619 | 1 h, 250 GB | brain02, 2026-11-03 |

**Not run (stateful).** left out by the table rules (ok): 5 correct bag, 5 early-reply bag, 6 correct bag.

skipped: a smaller size had reached its limit: 7 correct bag, 7 early-reply fifo, 7 early-reply bag, 7 correct fifo, 8 correct fifo, 8 early-reply bag, 8 early-reply fifo, 8 correct bag, 9 early-reply bag, 9 correct fifo, 9 early-reply fifo, 9 correct bag.

left out by the table rules (timeout): 6 early-reply bag, 6 early-reply fifo.

the tool ended with an error (exit-101): 7 correct bag, 7 early-reply bag, 8 early-reply bag, 8 correct bag, 9 correct bag.

planned but never scheduled (grid points beyond the ladder): 5 early-reply fifo, 5 correct fifo.

# offline

## offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 8 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.3 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 16 | 128 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 144 | 4608 | conforms | 0.7 | 4 | 10 min, 16 GB | 0.1 | 0.7 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 2304 | 294912 | conforms | 18.1 | 6 | 1 h, 250 GB | 0.5 | 18.0 | 0.1 | merge | brain05, 2026-11-03 |
| 5 | 57600 | 29491200 | conforms | 713.0 | 136 | 1 h, 250 GB | 5.1 | 711.5 | 1.5 | merge | brain05, 2026-11-03 |

### offline: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 8 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 4 | 128 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 36 | 4608 | conforms | 0.7 | 4 | 1 h, 250 GB | 0.1 | 0.7 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 576 | 294912 | conforms | 17.0 | 4 | 1 h, 250 GB | 0.2 | 16.9 | 0.1 | merge | brain05, 2026-11-03 |
| 5 | 14400 | 29491200 | conforms | 736.4 | 136 | 1 h, 250 GB | 1.8 | 734.9 | 1.5 | merge | brain05, 2026-11-03 |

### offline: pair **early-reply**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 32 | 3 | violation | 0.3 | 4 | 10 min, 16 GB | 0.3 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 16 | 512 | 12 | violation | 0.2 | 4 | 10 min, 16 GB | 0.2 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 144 | 18432 | 108 | violation | 1.8 | 4 | 1 h, 250 GB | 0.1 | 1.8 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 2304 | 1179648 | 1728 | violation | 41.5 | 10 | 1 h, 250 GB | 0.5 | 41.4 | 0.1 | merge | brain05, 2026-11-03 |
| 5 | 57600 | 117964800 | 43200 | violation | 2073.0 | 572 | 1 h, 250 GB | 5.1 | 2069.4 | 3.6 | merge | brain05, 2026-11-03 |

### offline: pair **early-reply**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 32 | 3 | violation | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 4 | 512 | 12 | violation | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 36 | 18432 | 108 | violation | 1.8 | 4 | 1 h, 250 GB | 0.1 | 1.8 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 576 | 1179648 | 1728 | violation | 42.1 | 14 | 1 h, 250 GB | 0.2 | 42.0 | 0.1 | merge | brain05, 2026-11-03 |
| 5 | 14400 | 117964800 | 43200 | violation | 1960.9 | 567 | 1 h, 250 GB | 1.8 | 1957.3 | 3.6 | merge | brain05, 2026-11-03 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 5 early-reply fifo, 5 early-reply bag, 5 correct bag, 5 correct fifo, 6 correct bag, 6 early-reply bag, 6 early-reply fifo, 7 correct bag.

skipped: a smaller size had reached its limit: 6 correct bag, 6 early-reply bag, 6 correct fifo, 6 early-reply fifo, 7 correct bag, 7 early-reply fifo, 7 early-reply bag, 7 correct fifo, 8 correct fifo, 8 early-reply bag, 8 early-reply fifo, 8 correct bag, 9 early-reply bag, 9 correct fifo, 9 early-reply fifo, 9 correct bag.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 8 | conforms | 3.4 | 185 | 10 min, 16 GB | 0.3 | 3.3 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 8 | conforms | 3.4 | 168 | 10 min, 16 GB | 0.1 | 3.3 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **early-reply**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 32 | 3 | violation | 3.4 | 154 | 10 min, 16 GB | 0.3 | 3.3 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **early-reply**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 32 | 3 | violation | 3.4 | 166 | 10 min, 16 GB | 0.1 | 3.4 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Java implementation against the Rust specification).** refused (an input missing at run time): 2 correct bag, 2 early-reply bag, 2 early-reply fifo, 2 correct fifo, 3 correct bag, 3 correct fifo, 3 early-reply bag, 3 early-reply fifo.

planned but never scheduled (grid points beyond the ladder): 5 early-reply bag, 5 correct bag, 6 correct bag, 6 early-reply bag, 7 correct bag.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 8 | conforms | 3.1 | 165 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 16 | 128 | conforms | 3.4 | 166 | 10 min, 16 GB | 3.4 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 144 | 4608 | conforms | 3.8 | 166 | 10 min, 16 GB | 3.8 | 0.7 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 8 | conforms | 0.7 | 160 | 10 min, 16 GB | 0.7 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 4 | 128 | conforms | 3.1 | 160 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 36 | 4608 | conforms | 3.9 | 158 | 10 min, 16 GB | 3.9 | 0.7 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Rust implementation against the Java specification: pair **early-reply**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 32 | 3 | violation | 3.1 | 165 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 16 | 512 | 12 | violation | 3.4 | 166 | 10 min, 16 GB | 3.4 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 144 | 18432 | 108 | violation | 3.8 | 166 | 10 min, 16 GB | 3.8 | 1.8 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Rust implementation against the Java specification: pair **early-reply**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 32 | 3 | violation | 0.7 | 160 | 10 min, 16 GB | 0.7 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 4 | 512 | 12 | violation | 3.1 | 160 | 10 min, 16 GB | 3.1 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 36 | 18432 | 108 | violation | 3.9 | 158 | 10 min, 16 GB | 3.9 | 1.8 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 5 early-reply bag, 5 correct bag, 6 correct bag, 6 early-reply bag, 7 correct bag.

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 8 | conforms | 0.7 | 152 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 2 | 16 | 128 | conforms | 1.5 | 249 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3 | 144 | 4608 | conforms | 33.9 | 325 | 1 h, 250 GB | | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 8 | conforms | 0.7 | 150 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 2 | 4 | 128 | conforms | 1.6 | 235 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3 | 36 | 4608 | conforms | 37.7 | 1008 | 1 h, 250 GB | | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **early-reply**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 32 | 24 | violation | 0.8 | 146 | 1 h, 250 GB | 115 | brain05, 2026-11-02 |
| 2 | 16 | 512 | 384 | violation | 3.4 | 285 | 1 h, 250 GB | 161 | brain05, 2026-11-02 |
| 3 | 144 | 18432 | 13824 | violation | 119.8 | 1296 | 1 h, 250 GB | 224 | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **early-reply**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 32 | 24 | violation | 0.9 | 163 | 1 h, 250 GB | 130 | brain05, 2026-11-02 |
| 2 | 4 | 512 | 384 | violation | 4.0 | 281 | 1 h, 250 GB | 201 | brain05, 2026-11-02 |
| 3 | 36 | 18432 | 13824 | violation | 149.4 | 418 | 1 h, 250 GB | 241 | brain05, 2026-11-02 |

**Not run (cross online, Java implementation (JMC) against the Rust specification export).** planned but never scheduled (grid points beyond the ladder): 1 correct bag, 1 early-reply bag, 1 correct fifo, 1 early-reply fifo, 2 correct bag, 2 correct fifo, 2 early-reply fifo, 2 early-reply bag, 3 correct bag, 3 correct fifo, 3 early-reply bag, 3 early-reply fifo, 5 early-reply bag, 5 correct bag, 6 correct bag, 6 early-reply bag, 7 correct bag.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 8 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 2 | 16 | 128 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 3 | 144 | 4608 | conforms | 0.6 | 4 | 1 h, 250 GB | | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 8 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 2 | 4 | 128 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 3 | 36 | 4608 | conforms | 0.7 | 4 | 1 h, 250 GB | | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **early-reply**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 4 | 32 | 24 | violation | 0.0 | 4 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 2 | 16 | 512 | 384 | violation | 0.1 | 2 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3 | 144 | 18432 | 13824 | violation | 2.0 | 4 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **early-reply**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 32 | 24 | violation | 0.0 | 4 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 2 | 4 | 512 | 384 | violation | 0.1 | 4 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3 | 36 | 18432 | 13824 | violation | 2.1 | 4 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** left out by the table rules (ok): 1 correct bag, 1 early-reply bag, 1 correct fifo, 1 early-reply fifo, 2 correct bag, 2 correct fifo, 2 early-reply fifo, 2 early-reply bag, 3 correct bag, 3 correct fifo, 3 early-reply bag, 3 early-reply fifo.

planned but never scheduled (grid points beyond the ladder): 5 early-reply bag, 5 correct bag, 6 correct bag, 6 early-reply bag, 7 correct bag.